

The Anatomy of Ultramarathon Performance

Ultramarathon performance is an enigma. It's elusive and hard to understand, and even when you are successful, you look back at the performance and wonder how exactly you accomplished what you did. Sometimes you don't even remember much of the ordeal, chalking the experience up to "ultra-amnesia" or some other concocted disorder. The difficulty of defining exactly what underpins ultramarathon performances shows up in the wide array of reports from runners at the end of an ultramarathon: "My legs were shot." "I got too hot on the exposed section." "My stomach went south." "I was really good on the climbs, but not so good on the descents." This array of experiences and associated tales have led ultrarunners and coaches to adopt a barrage of training strategies to combat those ailments. And, just like the event postmortems you'll hear at the finish line, basic training components like training volume, the proportion of time spent training at a certain intensity, supplemental training like strength training, and even coaching philosophies will vary wildly.

Even elite ultrarunners, who should be a more homogenous training cohort, show vast differences in training strategy. In 2018, I profiled *Ultrarunning Magazine's* "Ultrarunners of the Year," Jim Walmsley and Courtney Dauwalter. My aim was to uncover a bit of what made them successful that particular year. They both had excellent seasons with multiple wins across a wide variety of ultradistances, punctuated by wins at the famous Western States 100. Since I did not coach either of these athletes, I recreated their training from interviews I conducted, publicly available information on Strava, and other sources such as their social media accounts and prior interviews they had given. I've illustrated their training in Figures 3.1 and 3.2. Some aspects of their training, like annual training volume, were quite similar. (Note: Through the course of this book, I will refer to training volume in terms of hours per week as opposed to miles per week. This is because a time-based method of capturing volume is more consistent compared to a mile-based method. As any trail runner will know, the time and effort it takes to run a mile can vary wildly depending on if you are going uphill, downhill, or over technical terrain.) Courtney's 719 hours of annual training was quite comparable to Jim's 745 hours. They both used the principle of specificity and tailored the vertical gain and terrain of their training to a race as it approached. Yet other training variables were markedly different. For example, Jim's amount of vertical gain accumulated during the year (926,906 ft) nearly doubled Courtney's vertical gain (453,000 ft).

You see these types of discrepancies in training strategies throughout the pack. Some athletes will use high-volume strategies, running in excess of thirty hours per week, while other athletes will focus on intensity, doing three or four hard workouts per week. Some athletes swear by strength training, while others couldn't tell you the difference between a trap bar and a Snickers bar.

Some of this discrepancy is fueled by ultrarunning's history. Weird, obscure, and rooted in anecdote, training methods from ultrarunning's past have been indecipherably passed down from person to person in a decades-long game of telephone. The now defunct *Ultra-list*, a message board popular in the 1990s and early 2000s, was filled with endless iterations of "here's what I did" successes and failures from the early pioneers of ultrarunning. With each subsequent telling of the tale, the training pattern described moved further and further away from the truth and eventually resembled something closer to a fishing tale than a training log.

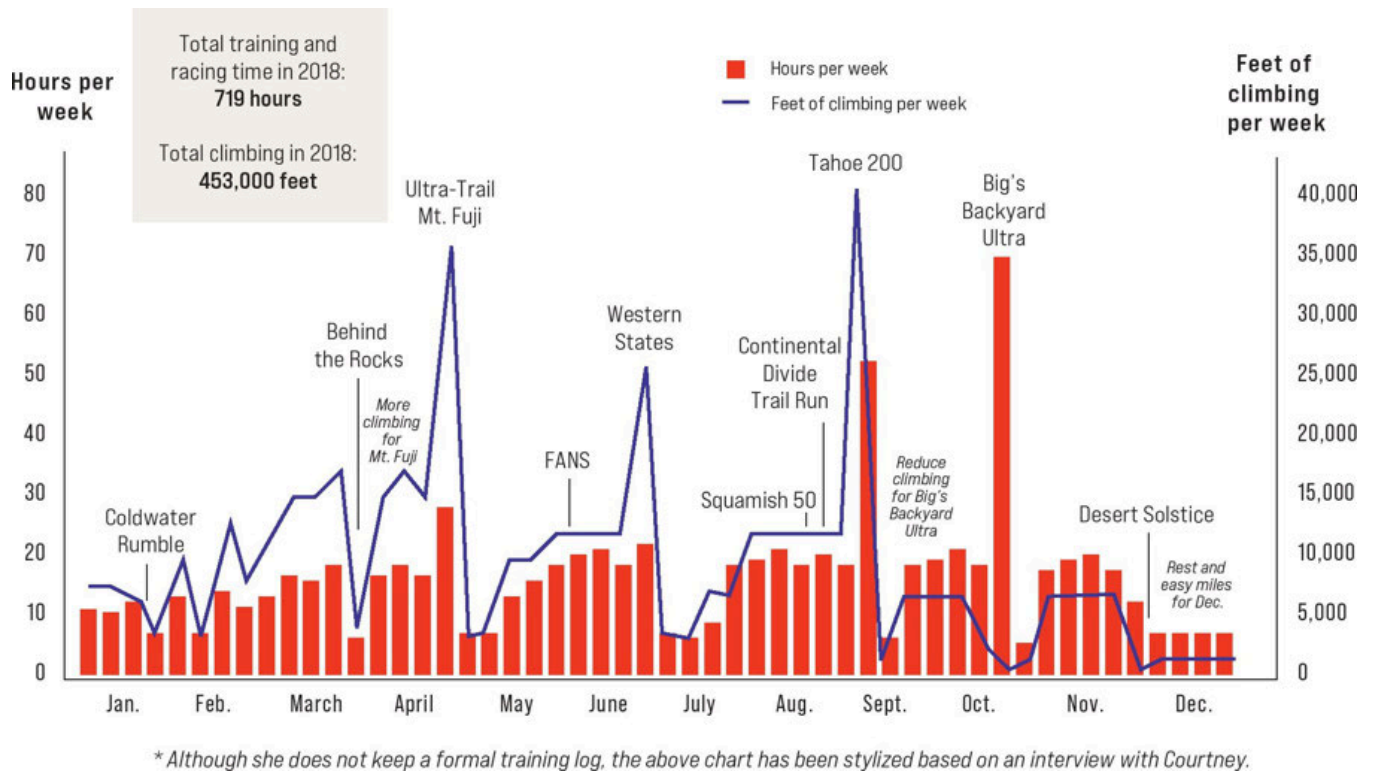


FIGURE 3.1 Courtney Dauwalter's 2018 training. Source: Koop 2019

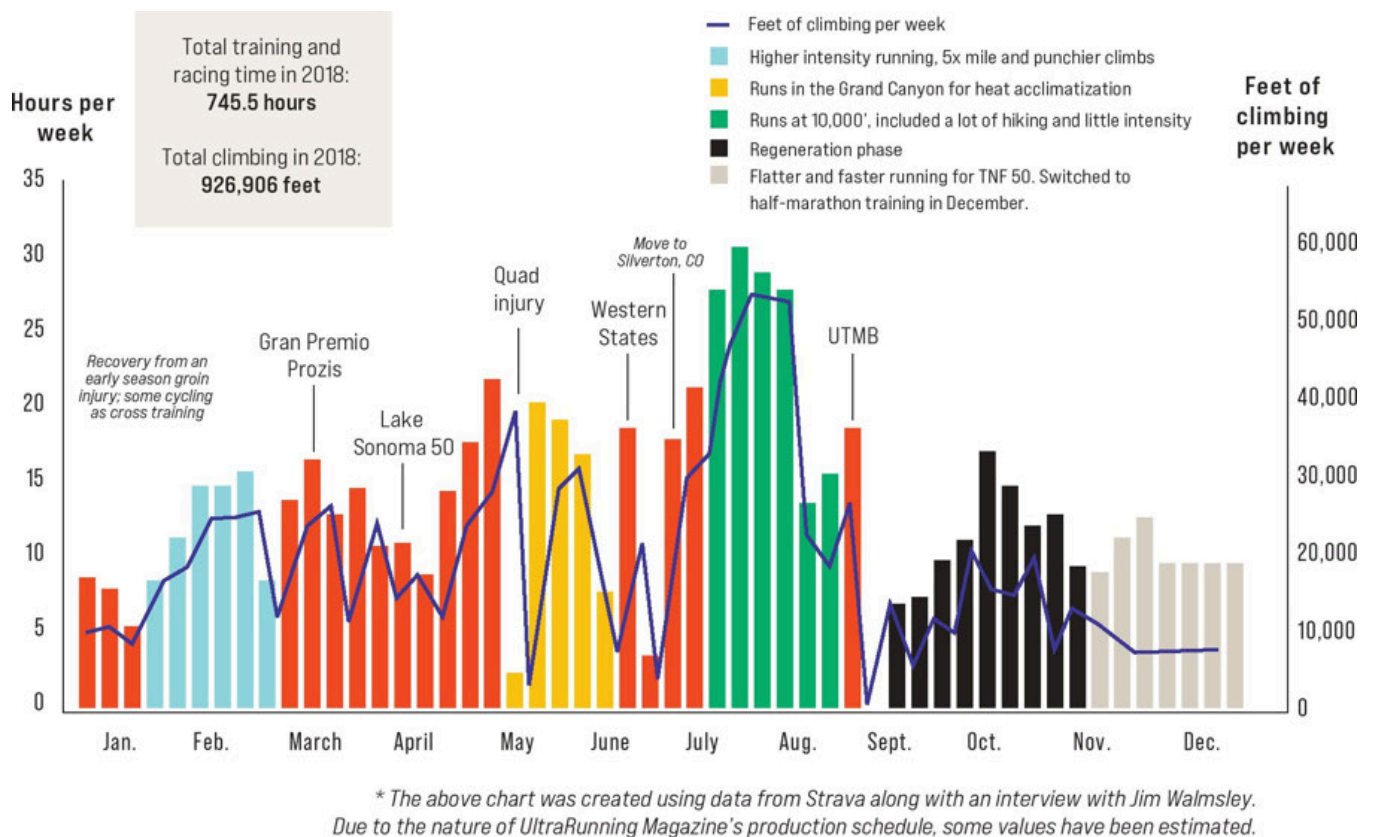


FIGURE 3.2 Jim Walmsley's 2018 training. Source: Koop 2019.

Other parts of these training discrepancies can be chalked up to our individual adaptations to training. “It worked for me,” while a pure anecdote, has legitimate physiological roots. According to the principle of individuality that I described in Chapter 2, different people respond to different types of training differently. Coaches have known this for decades through simple observation. Within a group of athletes given the exact same training program (like in a collegiate cross-country setting), individuals adapt differently to speedwork, high-volume training, and other training components. In more recent years, physiologists and coaches have attempted to tease out precise mechanisms that create these individual variations. They’ve determined that differences in muscle fiber type distribution (Bellinger et al. 2020; Leivens et al. 2020) and genetic coding (Bouchard 2012; Sarzynski, Ghosh, and Bouchard 2017) can influence how athletes adapt differently to a particular training protocol and even that our psychological response to training is different from one athlete to another (Wahl et al. 2013). Going down this rabbit hole, coaches have attempted to tailor training plans based on these narrow underpinnings, altering training for their “fast-twitch” vs “slow-twitch” and “tough” vs “not-so-tough” athletes. These narrowly focused attempts to individualize training ultimately miss the mark, as one single point of difference is unlikely to explain the whole cascade of physiological changes that result from training. It is the constellation of many different individual variances, not any single one of them, that ultimately constitutes the individual variability of a particular training protocol.

THE PROBLEM WITH ANECDOTE

WE’VE ALL HEARD TALES, TIPS, AND TRICKS from fellow ultrarunners on what has worked for them. Exclamations such as “Chew gum to alleviate a sour stomach—it worked for me,” “I drank some soup at mile 70 and it worked wonders,” and “These toe socks are the only socks that won’t give me blisters” abound in pre-race meetings, aid stations, and Facebook groups alike. While true and well-meaning, these anecdotal tales are a poor way to construct any ultramarathon plan. Why? First off, you can always find anecdotes both for and against any intervention. For every runner that magically turned their day around with a grilled cheese sandwich, there is another runner who threw it up immediately after leaving the aid station. So, is runner #1’s anecdote more valuable than runner #2’s? Furthermore, many anecdotes do not equal data. Just because your Facebook straw poll indicates that everyone is wearing HOKAs does not mean that they are the right shoe for you. You can find similar sentiments across a wide variety of interventions ranging from low-carbohydrate diets to what type of gel is best and even if an altitude tent works.

As coaches, we are trained (or at least we should be trained) to use a hierarchy of evidence when evaluating such claims. At the top of the food chain are systematic reviews and meta-analyses. These studies of studies take dozens and sometimes hundreds of similar research studies and filter through the findings to ascertain interventions that have worked and not worked in sports science. We can then use these types of interventions and have a reasonable degree of confidence that they are going to have some positive effect on the athlete. Adjunctive training like strength training, use of a sauna, block training, and altitude training (all of which are discussed in this book) fall into this category. They have all had systematic reviews and meta-analyses performed, and we can use those findings to inform training.

Next, we use randomized controlled trials and case-controlled studies. These singular studies look at the effects of a specific intervention across a homogenous group of people. For these two categories of research, we can take the findings and create applicable extensions by combining them with the fundamental aspects of physiology to create reasonable ways to train athletes.

Finally, on the hierarchy of evidence, we have anecdotes and field reports from athletes and coaches. These tales of “it worked for me” often become the basis for designing a randomized controlled trial or case-controlled study. However, they should not take precedence over information gleaned from randomized controlled trials, case-controlled studies, or meta-analyses. Anecdote can add context to those findings, but if you are going to supersede a finding from a meta-analysis with a tale from one of your running friends, you might as well admit that you are throwing a dart at a dartboard while blindfolded.

Cracking the Code of Ultramarathon Performance

When I first began coaching ultrarunners, I examined research-based models to try and identify the limiting factors for performance. In the marathon world, well-established research described performance being limited by three factors:

1. VO₂ max, or the maximum amount of oxygen an athlete can utilize in one minute of maximal exercise
2. The fraction of that VO₂ max an athlete can sustain over the course of the marathon
3. The oxygen cost of running, or how much oxygen it takes to go a given distance

Manipulating any or all of these three variables can influence marathon performance. This performance prediction was simple and robust enough that in 1991, Dr. Michael Joyner boldly predicted that the upper limit of marathon performance was 1:57:58 if one were to optimize all three of those variables (Joyner 1991). Fast-forward twenty-six years, and a team of scientists, product developers, and athletes decided to test the hypothesis. Led by Brad Wilkins, PhD, and the Nike Performance Laboratory, the team sought athletes with the greatest combination of the aforementioned physiological characteristics and then placed them in an environment with the equipment and conditions that would exploit those variables and maximize their chances of breaking the two-hour marathon threshold. They found a pool of over a dozen athletes who simultaneously possessed very high VO₂ max values, the ability to utilize a large fraction of their VO₂ max, and an extremely low cost of running (meaning that they required less oxygen to run a given distance).

Finding the athletes was not enough. After all, athletes race marathons all the time, and up to that point, no one had come close to breaking the two-hour mark. So, the team started to push and pull on other parts of the sub-two-hour equation. They researched the optimal temperature so the athletes could utilize a greater fraction of their VO₂ max for forward motion rather than dissipating heat. Advanced nutritional interventions were designed with precision, to be delivered to runners by bicycle in order to avoid any wasted motion. They sought the ideal conditions that would further drive down the cost of running by selecting a flat course at sea level that had very few turns. Teams of pacers would rotate to the front of the pack of runners to reduce the amount of aerodynamic drag, driving down the cost of running even further. As if that weren't enough, the entire entourage of athletes and their pacers ran behind a Tesla with a giant video board mounted on top in an attempt to squash what little wind resistance remained. Finally, Nike introduced the controversial Vaporfly shoes into the scenario, putting a final stamp on the cost of running reductions. I shouldn't need to tell you how this story ends.

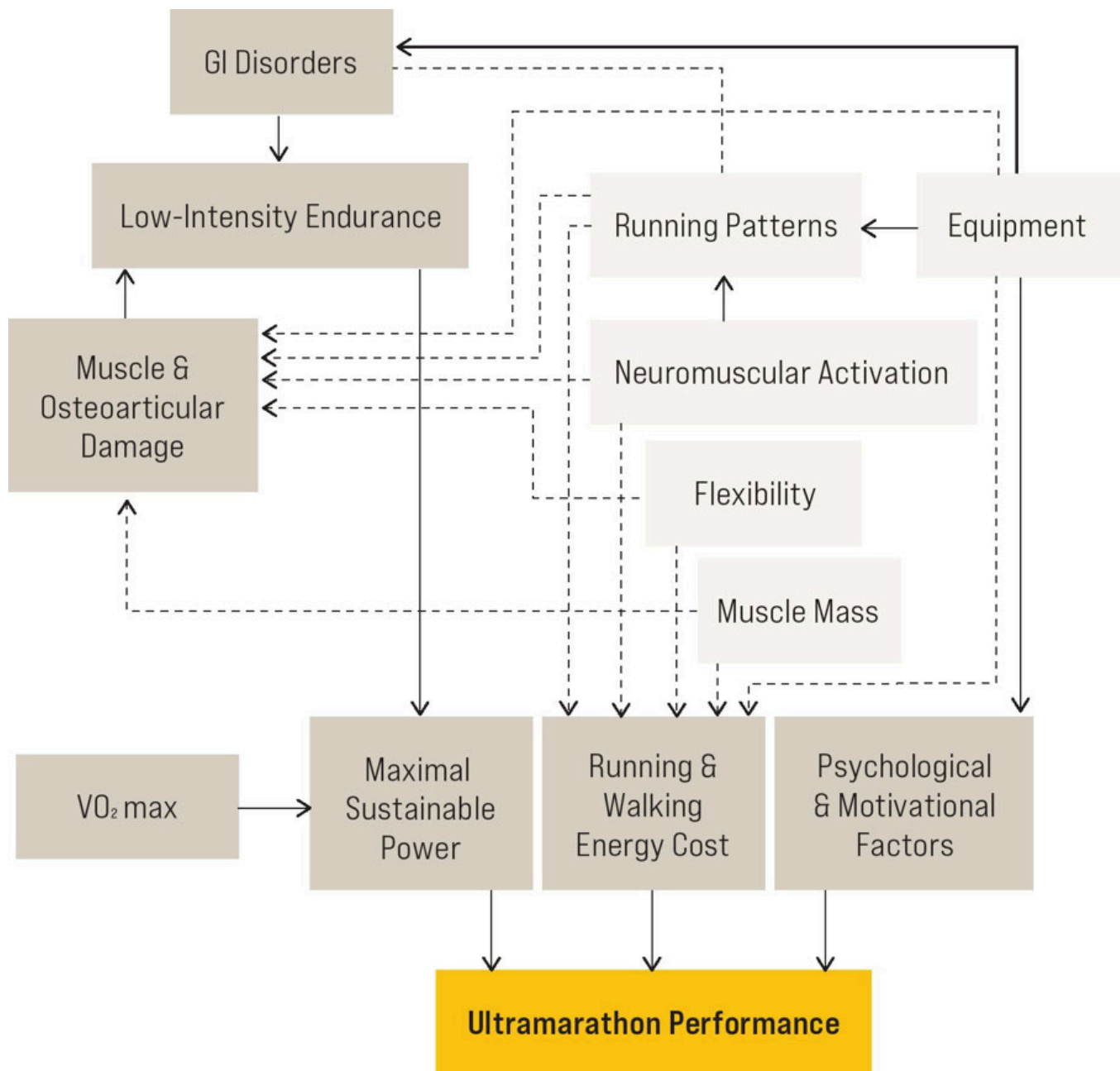


FIGURE 3.3 Proposed determinates of ultramarathon performance. Adapted from Millet 2012.

I would love to tell you ultrarunning is that simple. Push a little here, pull a little there, reduce this variable, and you'll magically run faster. Ultrarunning is not marathon running, and if I sound like a broken record repeating that basic fact throughout this book, it's for good reason. In fact, I will go on record right here and now and say that the marathon is likely the longest race where pure physiology is the main determinate of performance. In ultrarunning, physiology is still important, but your body will have to overcome markedly more obstacles that have greater potential for cataclysmic failure.

For what it's worth, researchers heavily invested in ultramarathon performance have started to make headway in finding the ingredients that make up the goulash of ultramarathon performance (Garbisu-Hualde and Santos-Concejero 2020; Knechtle and Nikolaidis 2018; Millet and Millet 2012, 2020; Millet et al. 2011; Thompson 2017; Knechtle and Senn 2011). And I have to say, while we've made a lot of progress in this area, we're a long way from a three-variable formula that describes the upper limits of ultramarathon performance (see Figure 3.3).

One way we can look at what optimizes performance is through the lens of the elite athletes who dominate the sport. Ultrarunning is now competitive enough, with a sufficient number of elite athletes, that we can look at that

landscape and discern general trends and traits that drive elite performance. Fortunately for you, I've had a front-row seat to that show over the past fifteen years. Here's what I've learned.

Characteristics of Elite Ultrarunners

Despite their diverse backgrounds, elite ultrarunners share three common characteristics. You have them, too, to varying degrees. Not only do you share these characteristics with elite runners, but all three can be developed by any runner to improve performance!

TALENT.

Elite ultrarunners (and elite endurance athletes in general) have a high genetic potential for aerobic power and other variables that constitute ultramarathon performance. This potential is trainable, so the takeaway for any runner is that no matter your starting point, working to optimize the talent you have will make you a better ultrarunner.

MENTAL SKILLS.

Elite ultrarunners possess supreme mental skills. They thrive not only on their best days, but at times on their worst days. The good news for all runners is that anyone can harness mental skills with focused training.

EMOTIONAL ENGAGEMENT.

Ultramarathons are hard. The best runners have an emotional attachment to the races they compete in; they care about the race and the community surrounding the event, not just their own performance. I encourage my athletes to pick races they have genuine, visceral attractions to, the type of attractions that makes you excited, giddy, and just a bit scared all at the same time. I would encourage you to do the same.

It is the harmonious intertwining of these qualities that ultimately enables elite ultrarunners to perform at their peak. While we all wish to run as fast as an elite athlete, reality dictates that most of us will be much slower. (Sorry to break the news.) The key is to optimize what you have by tapping into, leveraging, and developing your own innate talent, mental skills, and emotional engagement.

Talent

My formal education is in biochemistry and genetics. I spent the better part of five years toiling away in labs, behind pipettes, graduated cylinders, and beakers, studying gene expression and its role in the development of living organisms. Thus, it is with a bit of bias that I consider talent to mean your genetic predisposition for athletic

performance. All elite ultrarunners are born with a high degree of innate talent. This gift bestowed upon them by their parents is completely out of their control. Genetics sets some of the parameters and physiological limits of athletic potential. It plays a large role in determining the height of your highest-possible jump and the speed of your fastest-possible run, even if we could optimize everything else, like training, nutrition, and recovery.

For an endurance athlete, physiological talent is largely (but not exclusively) measured through the amount of oxygen one can utilize, represented as VO₂ max. This measure, the maximum volume of oxygen an athlete can consume, transport, and use, is usually expressed as milliliters of oxygen per kilogram of body weight per minute (ml/kg/min). Typical VO₂ max values for elite athletes in a variety of sports are shown in Figure 3.4. For endurance athletes, the rate of oxygen consumption is your most significant limiting factor, or your greatest advantage. The oxygen you take in is used to release energy through the breakdown of carbohydrate and fat (and, in dire circumstances, protein). The more oxygen you can utilize, the more rapidly you can deliver energy and, therefore, the greater the work capacity you have for an ultra.

To be an elite endurance athlete, you have to be able to utilize the oxygen required to keep you at the front of the pack. When an athlete has the necessary VO₂ max values, we typically say they are “in the club.” In long-distance running, for example, a typical elite male will have a VO₂ max around 77.4 ml/kg/min (Nevill et al. 2003). This does not mean that all elite runners have this level of maximum aerobic output, nor does it mean that one can’t be an elite runner with less than a 77.4 VO₂ max. Instead, it means that in order to be an elite male runner, you have to at least be close to that number. Someone who comes into a physiology lab and tests at only 40 ml/kg/min has no realistic chance of “getting into the club” in elite distance running.

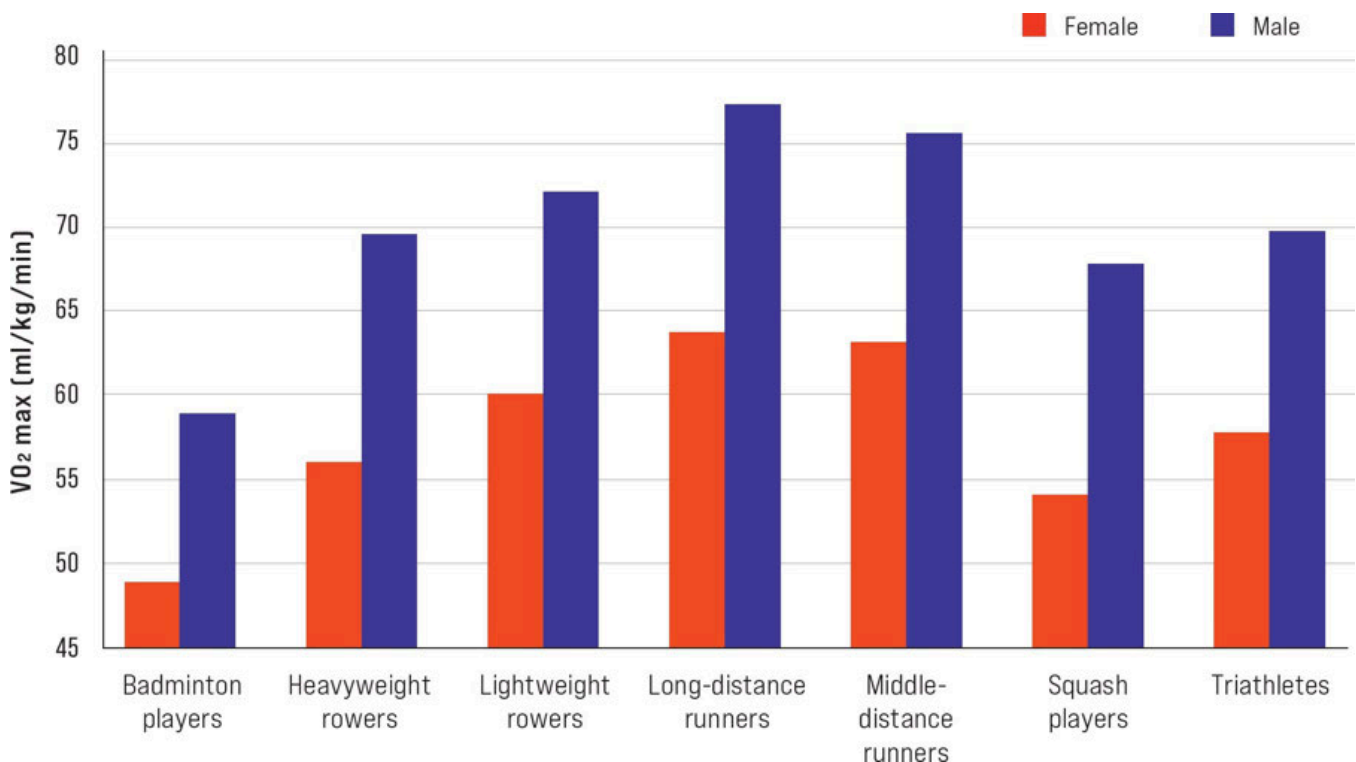


FIGURE 3.4 Typical VO₂ max values for elite athletes in various sports. Adapted from Nevill et al. 2003.

Elite ultrarunners don’t have to have VO₂ max values as high as elite marathoners, but they still have to be able to consume oxygen at or above a certain rate in order to compete at the top of the sport. In my testing of elite ultrarunners, successful males typically have VO₂ max values of approximately 60–70 ml/kg/min, and successful female competitors typically have VO₂ max values of approximately 55–65 ml/kg/min. This means simply that if you want to win a big, competitive race like the Western States 100, Leadville Trail 100, or Lake Sonoma 50, your VO₂ max values need to be in those neighborhoods.

In spite of how much emphasis I put on the aerobic system, I am the first to admit that an athlete's aerobic power is not the only physiological variable to consider. If races could be won simply by having a higher VO2 max, athletes could skip the running part and just show up with VO2 max test results to claim their prizes. To be a contender to win any major ultrarunning competition, an athlete needs a more diverse set of physiological talents compared to athletes in other running events. (I expand on this concept in Chapter 4.)

Elite ultrarunners are naturally skilled over varied terrain and equipped with good balance, reflexes, and an innate sense of where to put their feet. They have a talent for coping with environmental stressors and are able to consume food and fluid in great quantities in the middle of intense competition without getting sick. The fact of the matter is, elite ultrarunners are elites because they are talented across many physiological and psychological variables, not solely because of their aerobic power.

It is important to note that these aspects of talent are trainable. For novice runners, VO2 max can improve by well over 20 percent. Even an elite athlete with a very high VO2 max can still improve VO2 max ~5 percent with proper training (although for these athletes, VO2 max improvements are not nearly as important as improvements in other areas). Other aspects of innate talent can be similarly improved. You can train to consume more food, run over technical terrain, and handle hot and cold conditions. In these respects, all athletes are working to maximize their innate talents; elite athletes just have a higher ceiling.

WHY TALENT MATTERS

Very few athletes have the constellation of physical gifts required to be an elite ultrarunner, but talent is nevertheless important for all of us. Every athlete has some level of talent, and it is important to identify the areas where genetics and predispositions are well suited to ultrarunning. Maybe you don't have a world-class VO2 max, but you have a better-than-average ability to maintain your lactate threshold pace for long periods. This ability is useful for events with long climbs, as well as many other areas. Maybe you adapt remarkably well to high altitude, so your performance and pace don't drop as much at higher elevations. Or perhaps you are able to sustain a heavy training workload week after week because you recover and adapt to training stress quickly. These are all innate traits you can optimize and leverage in training and competition.

When elite ultrarunners line up at the start of a major race, they are very similar to one another in terms of talent. On paper it would appear to be a race among equals, yet the runners' real-world performances vary greatly on any given race day. Where these athletes differ is in how much they have closed the gap between their actual performance level and their maximum potential. Because the athletes have very similar levels of talent, those who can operate closest to their physiological ceiling have the advantage.

No matter where your own physiological ceiling is, the goal of training is to close the gap between your current performance level and your maximum potential. The great thing is that there is always room for improvement, because none of us, not even Olympic-caliber endurance athletes, operate at our maximum. We can get very, very close to that theoretical maximum, but we always come up a little short. That persistent gap is the reason we can all improve with proper training.

MENTAL SKILLS

Four to one is my typical rule of thumb: for every four workouts I give an elite athlete, I'm holding them back in some way on one of them. This is because elite athletes are naturally very mentally tough. All too often, they are too tough for their own good. They have a tendency to push through injury and illness, sometimes to the point where these problems become unnecessarily serious. This can be a bad thing in day-to-day training, but toughness is a golden quality on race day. Elite ultramarathoners train a lot. That's one of the bigger advantages they have over the rest of the pack. They organize their lives so they can train for many hours. However, even elite athletes can only prepare so much for race day. The goal is always to be 100 percent prepared, but reality dictates that they

will show up at the starting line with at least a few chinks in their armor. Maybe they left mileage on the table. Sometimes they couldn't get in enough vertical. In these instances, mental skills can provide strength in areas training left vulnerable. Elites reduce or eliminate these vulnerabilities better than the rest of the pack.

WHY MENTAL SKILLS MATTER

No matter how well you prepare for an ultra, it is nearly impossible to be fully ready for everything the race will throw at you. The elites have the mental skills to maintain a performance level that will keep them in contention for victory. Sometimes, when victory is no longer within reach, they have to rely on those skills just to keep moving forward. For a non-elite ultrarunner, mental skills may be even more important because you are out on the course longer, so there's more time to be affected by adversity, for unfavorable weather to creep in, and for your stomach to turn against you. While you're battling through fatigue at hour 23 or 27, the elites crossed the finish line hours earlier. Fortunately, mental skills can be developed and honed. They are forged through day-to-day training and through learning to endure specific challenges as you prepare for your ultramarathon event. You develop mental fortitude and grit by rising to those challenges, like pushing through difficult workouts and working through bad patches during long runs. This is such a crucial aspect of performance that I will spend an entire chapter on the subject (Chapter 15) later in this book.

EMOTIONAL ENGAGEMENT

Elite athletes have a high level of emotional engagement in the events they choose, and it shows. They have a tendency to care about the community surrounding the event, not solely their own performance on race day. During interviews, the elites demonstrate having a sense of history, past winners, and every rock, nook, and cranny of the racecourse. Many times, they know the aid station captains by name. And after they are done competing, they often volunteer for the races they have won and come to love.

In 2015, Western States 100 winner Rob Krar ran the final mile with the last finisher, in flip-flops. Think about that for a second. After running 100 miles and resting for about fourteen hours, this guy had the energy and emotional investment to escort the final finisher all the way to, and around, the Placer High School track. He could have easily sat in the bleachers with the rest of the crowd and perhaps mustered the energy to join the standing ovation. But he did so much more. He took his emotional engagement several hundred steps farther. He cared about the overall race, and he cared about that last finisher. He has an emotional engagement with the Western States 100 that surpasses his own planning, preparation, and the rigors of race day.

A large part of success in any elite competition is the athlete's emotional engagement with the event. We have all seen extremely talented, well-trained athletes underperform on the field of battle because they were "checked out." It happens in every sport and at every level of competition, even in elite ultrarunning. The elites I work with are able to go to almost any race on the planet. When I sit down with them at the beginning of every season, the number of opportunities on the table is unmanageable. During this process of picking and choosing events, it is easy to identify the characteristics of the races that would best suit the athlete's physical abilities. It is easy to look at a race and consider how much climbing and descending it includes, how hot or cold it will be, and whatever other variables exist and say, "Well, you are good at X, Y, and Z, so go do the races with X, Y, and Z." However, I always begin with finding events that have the most emotional pull for each athlete. I put my athletes in a position for success by first encouraging them to train for events they genuinely care about and then building their physical tools around that event, not the other way around.

WHY EMOTIONAL ENGAGEMENT MATTERS

For the last several years, I have been working with an athlete who has been attempting to complete the Leadville Trail 100 within the required thirty-hour cutoff. As of this book's publication, she unfortunately has been unsuccessful in this endeavor. Based on her innate talent, the thirty-hour cutoff for that race is within the limits of her physical capabilities, but only by a razor-thin margin. For her to be successful, everything has to go right. In training she has to make the most of every day, completing each workout to the fullest and resting with purpose. During the race she needs perfect weather and flawless race execution, and she has to dig into her training-honed well of mental skills deeper than ever before. If, and only if, all these things go right, she has a chance to be successful.

On her very best day in her very best year, this athlete is capable of a 29:45:00 finish. For the last four years she has been on the other side of that coin, yet she continues to go back, and I wholeheartedly encourage her to do so. I could easily coax her into an easier race with a more generous cutoff—perhaps a 100-miler at sea level. There are numerous events that would greatly increase her chances of finishing her first 100-miler, on paper at least. Yet, despite what I know from the rudimentary mathematical exercises of cutoffs, paces per mile, and probability ratios, I refuse to talk her out of racing Leadville. The sole reason for this is that she is 100 percent head over heels all-in infatuated and in love with the Leadville Trail 100. She is more emotionally engaged with that event than with any other race. So, even though on paper another event might be “easier” for her, I would argue that her best chance of success in a 100-mile footrace is in the race she's most passionate about.

Many people race in events they have no attachment to. I honestly don't know why. Even when you love the sport and the event you're preparing for, at some point you will want to quit. When you're exhausted, wet, cold, and nauseated, a part of your brain will tell you it's just not worth it, and you will quit. Training for and running an ultra are extremely hard. You'd better care about what you are doing.

TRAINING TIES TALENT, MENTAL SKILLS, AND EMOTIONAL ENGAGEMENT TOGETHER

Because this book centers on training for an ultramarathon, I'd be remiss if I neglected to tie together how training affects talent, mental skills, and emotional engagement. Quite simply, training is the catalyst that maximizes these qualities. You can improve by training more. You can improve by training harder. However, the most successful athletes also train smarter than their competitors. They use training to harmoniously maximize their innate talent, hone mental skills, and reinforce their emotional engagement for the event they are training for.

Training maximizes talent by pushing your raw physical capabilities ever closer to their predetermined genetic limits. Even rudimentary training moves you in the right direction, and training that is well designed further enhances your progress. The better, more intelligent, and more precise the training design, the closer you will get to your physical talent ceiling.

As for mental skills, their importance cannot be exaggerated. As much as I am an advocate for intelligent, precise training, I still want my athletes to work hard, push themselves when required, and be mindful of the work they have accomplished. After all, physical adaptation to training is Darwinian in nature. Training should be difficult because you need to impose enough stress in order to adapt. In this way, training hones your mental skills. The deliberate act of pushing yourself in training reinforces and builds the mental skills you will draw upon come race day.

Training is elective; with rare exceptions, we are not running to earn our next paycheck, which is all the more reason that emotional engagement is a key part of training. You choose to lace up your shoes, head out the door, and put in the miles. This ritual is a daily reminder of what you are ultimately training for and serves to reinforce the emotional engagement you have with your goals. Do you post the elevation profile of the race you are training for on your wall or refrigerator? That is a form of emotional engagement. The hills you run, the intensity of your efforts, and nearly all aspects of training should remind you of the event you have chosen to undertake.

KEY POINTS ON THE ANATOMY OF ULTRAMARATHON PERFORMANCE

Successful ultrarunners have used a variety of different training pathways to success.

While marathon performance can be described by VO2 max, the fraction of that VO2 max you can utilize, and the oxygen cost of running, ultramarathon performance depends on a broader array of variables.

Elite ultrarunners exhibit three distinct characteristics:

- **High levels of genetic talent (physical, psychological, and otherwise)**
- **Highly developed mental skills**
- **Strong emotional engagement with the events they have chosen**

All athletes can leverage these three attributes to better their ultramarathon performance.

Training ties talent, mental skills, and emotional engagement together.



